Problem 1: ideal flow coefficients

From the lecture appendix, with ideal contraction and velocity coefficients $C_c = C_v = 1$,

$$A_c = \frac{\pi D^2}{4}, \quad A_o = \frac{\pi d^2}{4}, \quad K_{o,\text{ideal}} = \frac{1}{\sqrt{1-(A_o/A_c)^2}}, \quad K_{\text{ideal}} = K_{o,\text{ideal}} A_o \sqrt{\frac{2g}{A_c}}.$$

For Can Set 1, use $g = 981\,\text{cm/s}^2$ and $1\,\text{in} = 2.54\,\text{cm}$.

Quantity	Top can (1)	Bottom can (2)
D (cm)	10.17270	7.63778
d (cm)	0.63246	0.31242
A_c (cm^2)	81.2760	45.8167
A_o (cm^2)	0.314164	0.076660

$$\boxed{K_1 = 1.543574\ \text{cm}^{3/2}/\text{s}}, \qquad \boxed{K_2 = 0.501655\ \text{cm}^{3/2}/\text{s}}.$$

These are theoretical values for free discharge; the experimental coefficients can differ.

Problem 2: sample simulation

The supplied fourth-order Runge–Kutta code models

$$\dot{V}_1 = -K_1\sqrt{\max(V_1,0)}, \qquad \dot{V}_2 = K_1\sqrt{\max(V_1,0)} - K_2\sqrt{\max(V_2,0)}.$$

For $V_1(0) = 406\,\text{mL}$ and $V_2(0) = 0$, the predicted bottom-can peak is approximately 255.4 mL at 19.38 s, corresponding to a depth of 5.57 cm.

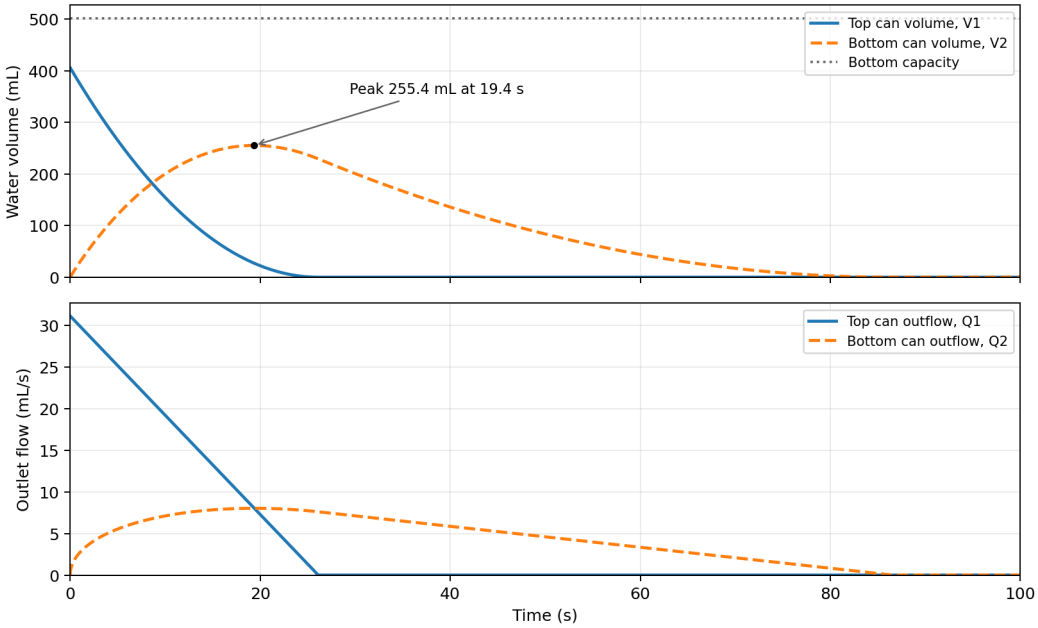


Figure 1: Simulated water volumes and outlet flow rates for Can Set 1 using ideal coefficients.

Problem 3: one-can experimental procedure

Objective. Determine K_1 and K_2 separately from emptying experiments, without electronic sensors. For a single can with no inflow,

$$\dot{V} = -K\sqrt{V}, \qquad 2\sqrt{V_0} = KT_e, \qquad K = \frac{2\sqrt{V_0}}{T_e}.$$

1. Set one can upright over a catch reservoir with its bottom orifice temporarily plugged. Keep the can level and ensure the discharge is unobstructed.
2. Measure or confirm its inside diameter D and compute $A_c = \pi D^2/4$. Mark the water heights corresponding to approximately 20%, 50%, and 80% of the can's usable height; calculate $V_0 = A_c h_0$ for each mark.
3. Fill to the first marked height. Record the initial water height h_0 and volume V_0; use the same release technique for every trial.
4. Remove the plug while starting a handheld stopwatch. Stop timing when continuous discharge ends (record the endpoint definition consistently). Record emptying time T_e.
5. Repeat each starting height three times, refilling between trials. Record can ID, D, h_0, V_0, T_e, and any observations (leaks, splashing, delayed release).
6. Repeat Steps 1–5 for the other can, collecting nine trials per can (18 total).
7. Compute $K = 2\sqrt{V_0}/T_e$ for each trial. Plot $2\sqrt{V_0}$ versus T_e and estimate K from the slope of a linear fit. Inspect any nonzero intercept as evidence of timing bias or model mismatch.
8. Report each can's fitted K, trial-to-trial variation, and the difference from its theoretical ideal value. Use the experimental coefficients in the two-can simulation for the in-lab prediction.

Measurement table columns: can ID, trial, D (cm), h_0 (cm), V_0 (mL), T_e (s), K ($\text{cm}^{3/2}/\text{s}$), notes.